

Mechanical Engineering junior specializing in SolidWorks CAD, DFMA, GD&T drawings, and structural FEA. Experienced in prototyping and testing mechanical assemblies, evaluating design changes

EDUCATION

University of Massachusetts, Amherst

August 2023 – May 2027

Bachelor of Science in Mechanical Engineering

Amherst, MA

- CGPA: 3.75/4 | Dean's List: F2023, S2024, F2024, S2025, F2025
- Relevant Courses: Design of Mech. Components & Assem., Dynamic Systems, Manuf. Processes, Prod. Planning & Control

EXPERIENCE

Robotics Design Engineer

September 2025 – Present

Robotics Research Club

Amherst, MA

- Designed modular leg assemblies for a quadruped robot by modeling linkages, joint housings, and torque-carrying interfaces in SolidWorks, improving load distribution and enabling reliable operation under expected joint torques.
- Developed complete URDF model, allowing the software team to run simulations with 1:1 joint and link geometry.
- Supported R&D through testing by iterating 3 prototype legs using 3D printing and FEA-driven refinements, validating range of motion, torque transfer consistency, and overall assembly fit for system-level trials.

R&D Engineer

March 2025 – July 2025

FootBit

Remote

- Modeled and refined compact enclosures for a wearable gait-analysis device in SolidWorks, achieving 28–30% dimensional reduction while maintaining wall-thickness requirements and assembly clearances through tolerance stack-up checks.
- Evaluated polymer housing options by comparing flexural rigidity, thermal behavior, and manufacturability, and used basic static FEA in ANSYS to verify deformation limits and safe heat buildup around the chip during use.
- Built and tested early prototypes to evaluate assembly robustness repeated-use loading, and produced manufacturing-ready drawings with GD&T for vendors to ensure accurate alignment features consistent for large-scale production.

Mechanical Engineer

June 2024 – August 2024

Interlock Group

Singapore

- Designed the mechanical framework for upper arm and joint connection for a 6 DOF warehouse robot in SolidWorks, incorporating DFMA-driven features into geometry and flange patterns to manage torque transfer effectively.
- Reduced assembly complexity by decreasing fasteners by 20% & consolidating connection plates into one single component
- Completed torque path analysis and bending/shear stress calculations for joint-link interfaces under up to 17–22 Nm motor output, ensuring a safety factor of 1.8 with aluminum 6061–T6 components.

Mechanical Systems Engineer

August 2023 – Present

UMass Aeronautics Team

Amherst, MA

- Designed and fabricated carbon fiber structural spars rated for 4G loading conditions, achieving 0.75lb weight reduction while maintaining stiffness through ANSYS FEA optimization and parametric MATLAB analysis.
- Engineered hatch and pylon attachment systems, minimizing mission time through systematic POF analysis of the wing.
- Executed precision CFRP fuselage fabrication and integration, maintaining critical structural tolerances for competition.
- Implemented Pugh Matrix analysis for material & manufacturing process selection, documenting in technical report.

Director

March 2025 – Present

HackUMass

Amherst, MA

- Coordinated operations for a 900+ participant hardware & software hackathon, managing logistics and tech supports.
- Expanded access to fabrication resources by partnering with makerspaces, enabling 100+ students to prototype hardware.
- Directed engineering outreach initiatives, fostering collaboration and increasing engineering participants by 100+ students.
- Directed engineering outreach initiatives that increased hardware-track engagement by 30–40% and led workshop on CAD.

PROJECTS

Undergraduate Researcher | *Manufacturing & Automation Lab*

September 2024 – December 2024

- Mechanically and electrically integrated the Robotiq vision camera with UR5e in the Manufacturing & Automation lab.
- Developed framework for MIE 375 enabling unstructured palletization operations with detailed process documentation.
- Optimized motion planning and trajectory algorithms in ROS2 simulation environment, reducing assembly time by 50%.
- Learned PLC coding and solved design and manufacturing challenges, improving system reliability and workflow efficiency

TECHNICAL SKILLS

Design & Analysis: SolidWorks, CATIA, Fusion 360, Onshape, MATLAB SimuLink, ANSYS FEA
Programming: Python, C++, MATLAB, ROS2, PLC Programming, Embedded Systems (ESP32)
Manufacturing: 3D printing, Wood cutting, Laser Cutting, Sheet Metal manufacturing, CNC milling